

Wireless Communication (EX751 / EX715)

Concise PYQ

July 22, 2026

Notes: TU IOE PYQ 2070–2081. Regular = **bold**, Back = normal; BEX = normal font, BEI (EX715) = **mono**. Marks in (); unique values only. Keys: (I)=Importance/Need (+)=Advantage (-)=Disadvantage +Fig=Figure +Eg=Example Vs=Compare/Differentiate.

1. Introduction

- Evolution of wireless / cellular (1G–5G); technology + market penetration (4)(6)
- 1G/2G/3G/4G Vs, technology advancement; 5G Vs 4G; forward & reverse channel (4)(6)(2+2)

2. Cellular Concepts

- Frequency reuse; co-channel reuse ratio $Q=\sqrt{3N}$ (derive); find N (omni/120°/60°) (6)(4)(1+4)(3+5)
- Min. co-channel distance (7-cell); co- Vs adjacent-channel interference (4)(3+4+3)
- Handoff — margin +Fig, strategies (GSM), types & application, cell dragging (3)(6)(8)
- GoS + trunking (blocked-call-cleared); market penetration; clusters/capacity (3+5)(5)(12)(6)
- Microcell zone concept; sectoring Vs cell splitting; capacity/coverage enhancement (3)(5)(8)

3. Radio Wave Propagation

- Free-space & link budget — received power / E-field / flux / rms voltage (8)(6)
 - TX/RX distance + gains (50 W@900 MHz; 165 W@325 MHz; 150 W threshold; monopole 2-ray)
- Propagation mechanisms; large- Vs small-scale; define path loss + multipath params (3+5)(8)(2+4)
- Two-ray / Fresnel diffraction — phase difference; scattering + ground-reflection model (8)(2+6)(2+8)
- Outdoor models — Okumura / Hata (median path loss, correction factor, EIRP) (4+4)(4+2+2)(3+3)
 - Ericsson Multiple Breakpoint model; 10-km link feasibility; 2-ray point-to-point (2)(12)(4+4)
- Indoor models; small-scale fading types + factors (8)(2+4+4)(4+4)
- Doppler spread / coherence BW & time; delay spread; T_s without equalizer (4+4)(4)(8)
- Rayleigh Vs Rician fading; flat Vs frequency-selective (AMPS/GSM) (2)(5)(2+6)

4. Modulation–Demodulation

- QPSK / OQPSK / $\pi/4$ -QPSK — Tx & detection; DPSK + PN sequence (6)(8)(5+2)(4+2+2)

- MSK Vs GMSK Vs OQPSK (BW/power efficiency, GSM); GMSK Tx + constellation (3+3+2)(2+6)(8)
- OFDM operation +Fig (Tx/Rx); modulation/demodulation technique (8)(4+4)
- Spread spectrum — DS & FH; Vs multiple-access techniques (4)(6+2)

5. Equalization & Diversity

- Equalization (I), minimize ISI; MSE (optimal weights); MLSE +Fig; adaptive algorithms (2+6)(8)(6+1+5)(4)
- Diversity (I) + types; combining (max-ratio / space); antenna diversity Vs (2+6)(2+2)(4+6)(3+3+2)
- RAKE receiver (M-branch) +Fig (8)(4)

6. Speech & Channel Coding

- Speech signal characteristics; vocoder (I) + predictive coders; GSM CODEC +Fig (8)(3)(2+6)(4+2+2)
- LPC + block diagram; sub-band coding (Tx/Rx); source coders (2+6)(3+5)(6)
- Convolutional / turbo coding; Viterbi decoding; interleaving (need + working) (5)(3)(4)

7. Multiple Access

- FDMA — non-linear effect, no. of channels; (+)(-); FDMA Vs CDMA (3+2)(4)(2+4)
- TDMA — (+) over FDMA; GSM frame efficiency (4)
- CDMA — characteristics/(+)(-); design considerations; Hadamard H_s (Walsh) (2+6)(4)
- SSMA (DS & FH); FHMA +block diagram +Eg; near-far effect + solution (4+4)(6)(7)(3+4)
- Hybrid SS MA (near-far mitigation); SDMA; multiple access Vs (TDMA/CDMA/SDMA) (2+6)(4)(2+6)
- WLAN; WiMAX (3)

8. Wireless Systems & Standards

- GSM architecture (each component); traffic & control channels; BCH & CCCH (8)(4+4)
- GSM frame structure; TDMA frame efficiency < 100%; speech-to-radio signal processing (6+2)(4+4)(8)
- CDMA IS-95 forward channels (draw); CDMA Vs LTE architecture (4)(3+5)
- Regulatory issues / spectrum regulation; interference; LTE / WiMAX / UMTS (4)(5)(3)